

FISAT GATEWAY MANAGEMENT SYSTEM

I. Modules :

>> Visitor Management Module

- Features: Record visitor details, view visitor logs, generate visitor passes.

>> Hostel Student Management Module

- Features: Record in-and-out details for men's and women's hostels, notify watchman and parents.
- Specialty: Automated SMS/Email notifications for permissions and updates.

>> Daily Reporting Module

- Features: Watchman records daily activities, admin reviews reports.
- Specialty: Report analytics with charts (e.g., number of visitors, staff, students per day).

>> Staff Movement Tracking Module

- Features: Log staff movements using ID card swiping via React Native.
- Specialty: Automatic entry using NFC integration.

>> Admin Module

- Features: View reports, manage user access, monitor all records.
- Specialty: Role-based access control and dashboard analytics.

II. Data Table:

>> Visitor Management Collection

Field Name	Data Type	Description
<code>_id</code>	ObjectId	Unique identifier for each visitor.
<code>name</code>	String	Visitor's name.
<code>purpose</code>	String	Purpose of the visit.
<code>timeIn</code>	Date	Time the visitor entered.
<code>timeOut</code>	Date	Time the visitor exited.
<code>contact</code>	String	Visitor's contact number.
<code>status</code>	String	Current status (e.g., <code>in</code> , <code>out</code>).

>> Daily Reporting Collection

Field Name	Data Type	Description
<code>_id</code>	ObjectId	Unique identifier for each report.
<code>date</code>	Date	Date of the report.
<code>visitorsCount</code>	Number	Total number of visitors for the day.
<code>studentsCount</code>	Number	Total number of students who moved.
<code>staffMovements</code>	Number	Total number of staff movements.
<code>remarks</code>	String	Additional notes or remarks.

>> Hostel Student Management Collection

Field Name	Data Type	Description
<code>_id</code>	ObjectId	Unique identifier for each record.
<code>studentId</code>	String	Unique ID of the student.
<code>name</code>	String	Student's name.
<code>department</code>	String	Department of the student.
<code>hostelType</code>	String	Type of hostel (<code>men</code> or <code>women</code>).
<code>timeIn</code>	Date	Time the student entered.
<code>timeOut</code>	Date	Time the student exited.
<code>parentContact</code>	String	Parent's contact number for notifications.
<code>wardenApproved</code>	Boolean	Approval status from the warden.

>> Admin Collection

Field Name	Data Type	Description
_id	ObjectId	Unique identifier for each admin.
username	String	Admin's username.
password	String	Hashed password for the admin.

>> Staff Movements Collection

Field Name	Data Type	Description
_id	ObjectId	Unique identifier for each record.
staffId	String	Unique ID of the staff.
name	String	Staff member's name.
timeIn	Date	Time the staff entered.
timeOut	Date	Time the staff exited.

II. Objectives:

>> Digitize Record Management

Replace the existing manual system of maintaining physical record books with a digital system for efficient and secure management of visitor, student, and staff movement records.

>> Enhance Data Accuracy and Accessibility

Minimize errors caused by manual entries and provide instant access to accurate records for administrators, wardens, and other authorized users.

>> Improve Communication

Facilitate real-time notifications to parents and watchmen regarding

student movements, ensuring transparency and better communication.

>> **Streamline Hostel Management**

Enable wardens to digitally log student entry and exit details, reducing dependency on physical slips and enhancing the approval process.

>> **Enable Centralized Reporting**

Provide a daily reporting feature for watchmen to summarize visitors, student movements, and staff exits, simplifying administrative oversight.

>> **Enhance Security and Monitoring**

Track all staff movements through a mobile application integrated with ID card swiping, ensuring better accountability and security on campus.

>> **Ensure Role-Based Access Control**

Implement a role-based system to restrict access to sensitive information, ensuring data security for wardens, watchmen, and administrators.

>> **Generate Comprehensive Reports**

Allow administrators to view detailed reports of daily activities, including student movements, visitor logs, and staff exits, for better decision-making and compliance.

>> **Ensure Scalability and Usability**

Build the system using modern technologies (MERN stack and React Native) to ensure scalability, cross-platform compatibility, and user-friendly interfaces for web and mobile applications.

>> **Support Eco-Friendly Practices**

Reduce the usage of paper by transitioning to a fully digital system,

contributing to environmentally sustainable practices within the campus.

III. Problems :

>> Time-Consuming Processes

- Logging visitor details, student movements, and staff exits manually in books takes significant time and effort.
- Retrieving old records requires manually searching through physical books, which is inefficient.

>> Human Errors in Data Entry

- Manual entries are prone to errors, such as incorrect or incomplete information.
- Misplaced or illegible records due to handwriting issues can lead to confusion.

>> Lack of Real-Time Communication

- Delays in informing the watchman about warden approvals or notifying parents of student movements.
- No system to ensure immediate communication or record verification.

>> Limited Data Security

- Physical books are vulnerable to damage, loss, or tampering.
- Sensitive information is not secured, leading to privacy concerns.

>> Inefficient Report Generation

- The watchman manually creates daily reports, which is a repetitive and error-prone task.
- Consolidating data for administrative purposes becomes cumbersome.

>> Dependency on Paper Slips

- Students rely on physical slips for exit permissions, which can be lost, misplaced, or forged.
- The manual slip system is inefficient and lacks verification features.

>> Inadequate Tracking of Staff Movements

- The punching machine used for staff exit tracking is outdated and does not integrate with a centralized system.
- No digital record exists to track or review staff movements effectively.

>> Difficulty in Data Analysis

- Manual records make it difficult to analyze trends or generate insights, such as peak visitor times or frequent student exits.
- Lack of digitization hampers decision-making and performance evaluations.

>> Environmental Impact

- The use of paper for maintaining records, slips, and reports contributes to wastage and is not eco-friendly.

>> No Centralized System

- Records are scattered across multiple books (visitor log, men's hostel, women's hostel), making it difficult to maintain a centralized view of activities on campus.

>> Lack of Accountability

- Manual records do not offer role-based access or a proper audit trail, making it hard to hold individuals accountable for discrepancies.

>> Scalability Issues

- The current system cannot handle a growing number of visitors, students, and staff efficiently.
- Scaling the manual process to meet future demands would require more manpower and resources.

IV. Ideas in Project :

>> Integration with Messaging Systems

Automated message systems to notify concerned parties (parents, watchmen) about significant activities, such as student movements or visitor approvals.

>> Real-Time Digital Notifications

- For Watchmen: Instant notifications to watchmen about warden-approved student movements, eliminating delays caused by paper slips.
- For Parents: Automated SMS or email notifications to parents whenever a student exits the campus, improving transparency and communication.

>> **Digital Approval Workflow for Hostels**

- A streamlined workflow for wardens to log student exit permissions digitally, complete with timestamps and detailed records.
- Removes dependency on physical slips and prevents loss or forgery.

V. Reference :

>> "Visitor Management System Design and Implementation during the Covid-19 Pandemic"

- *Authors:* Mohammed A. Al-Kabi, et al.
- *Source:* ResearchGate
- *Summary:* This paper discusses the design and implementation of a Visitor Management System (VMS) aimed at synchron

>> "Design and Implementation of Hostel Management System"

- *Authors:* J. A. Shittu, et al.
- *Source:* International Journal of Computer Applications
- *Summary:* Explores the implementation of a digital solution to manage hostel operations effectively.

